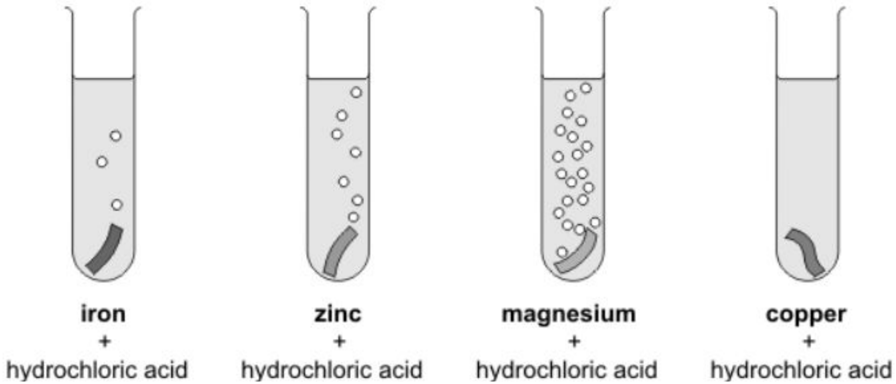
She poured 10 cm^3 of hydrochloric acid onto each metal.



Look at the diagrams above.

(i) How do these show if a metal reacts with the acid?

....

(ii) **On the lines below**, put the **four** metals in the order of how strongly they react with the acid.

most reactive ....

....

....

least reactive ....

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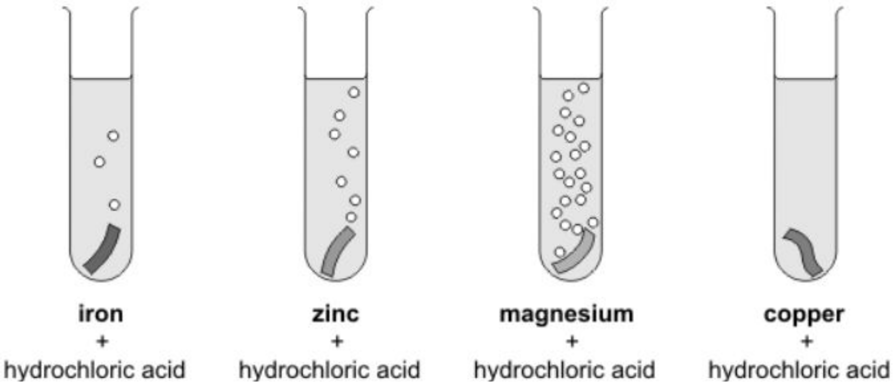

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Answers:

)22

(i) any **one** from

- bubbles
- fizzing

  accept 'effervescence'
- gas is given off

  'metal goes into solution ***or*** *turns into a salt' and 'there would be a rise in temperature' are insufficient answers as they are* ***not*** *shown in the drawings*

(ii)
- magnesium

  accept 'Mg'
- zinc

  accept 'Zn'
- iron

  accept 'Fe'
- copper

  accept 'Cu'

  answers must be in the correct order

  all four *answers are required for the mark*

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DO NOW

Displacement reactions:

Learning Objective

- Represent and explain displacement reactions using formulas and equations.
- Make inferences about reactivity from displacement reactions.

The reactivity series:

potassium	most reactive	K
sodium		Na
calcium		Ca
magnesium		Mg
aluminium		Al
carbon		C
zinc		Zn
iron		Fe
tin		Sn
lead		Pb
hydrogen		H
copper		Cu
silver		Ag
gold		Au
platinum	least reactive	Pt

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My question to you…

We have already found out that elements that chemically react can make a compound.

But..

What happens if this compound reacts with a new element?

<u>In your tables, discuss this and come up with a collective answer</u>

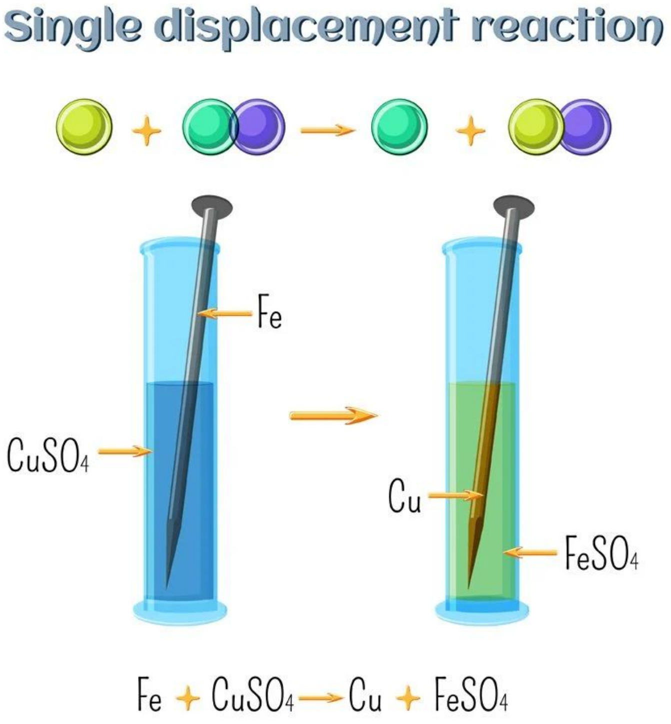
The answer….

It depends on how reactive the metals are!

Copper + Sulphuric Acid > Copper Sulphate

Displacement reactions:

- **In displacement reactions a more reactive metal displaces a less reactive metal in a compound**



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Displacement reactions

- What happens if…
- Copper Sulphate + Magnesium?
- How do we know whether it will be knocked off or not?

Potassium
Sodium
Calcium
Magnesium
Zinc
Iron
Copper
Silver
Gold

We can use our reactivity series to see if the reaction will take place

Displacement reactions:

Which is the **compound** in the following equation?

Copper sulfate + zinc → copper + zinc sulfate

What has happened in the equation?

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Displacement reactions:

Will a reaction be observed in the following:

Copper + zinc sulfate → copper sulfate + zinc

Hint: which metal is the most reactive?

Displacement r_______ involve a metal and a c_______ of a different m__. In a d____________ reaction a more r______ metal will displace a l__ reactive metal f___ it's compound.

displacement, from, compound, reactions, reactive, metal, less

Displacement r_______ involve a metal and a c_______ of a different m__. In a d____________ reaction a more r______ metal will displace a l__ reactive metal f___ it's compound.

displacement, from, compound, reactions, reactive, metal, less

Displacement reactions involve a metal and a compound of a different metal. In a displacement reaction a more reactive metal will displace a less reactive metal from it's compound.

I do

Displacement reactions:

What will happen in this reaction?

- Magnesium + zinc sulfate →

A: Magnesium + zinc sulfate → magnesium sulfate + zinc

We do

Displacement reactions:

What will happen in this reaction?

- Tin nitrate + magnesium →

A: Tin nitrate + magnesium → magnesium nitrate + tin

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You do

Displacement reactions:

Complete the following work equation:

1. Magnesium + silver nitrate → .... + silver

2. Aluminium + ... → aluminium sulfate + copper

3. .... + copper sulfate → zinc sulfate + ...

You do

Answers:

Complete the following work equation:

1. Magnesium + silver nitrate → magnesium nitrate + silver
2. Aluminium + copper sulfate → aluminium sulfate + copper
3. zinc + copper sulfate → zinc sulfate + copper

Displacement Reactions

Use the reactivity series to help you complete the word equations below.

Metal	Symbol
Potassium	
Sodium	
Lithium	
Barium	
Strontium	
Calcium	
Magnesium	
Aluminium	
Manganese	
Zinc	
Chromium	
Iron	
Cadmium	
Cobalt	
Nickel	
Tin	
Lead	
Copper	
Silver	
Platinum	

Most Reactive

Least Reactive

Calcium Oxide + Barium → __________ + ____________	Sodium Chloride + Potassium → _______________ + ______
Manganese Sulphate + Calcium → __________ + ___________	Iron Nitrate + Zinc → __________ + ____________
Tin Oxide + Aluminium → __________ + ____________	Lithium Oxide + Sodium → __________ + ____________
Cadmium Hydroxide + Aluminium → __________ + ____________	Silver Oxide + Copper → __________ + ____________
Strontium Sulphate + Potassium → __________ + ____________	Cobalt Chloride + Iron → __________ + ____________
Tin Sulphate + Zinc → __________ + ____________	Platinum Chloride + Silver → __________ + ____________

Colour all the reactions below that are not possible in green.

Zinc Oxide + Tin	Barium Sulphate + Lithium	Cobalt Chloride + Silver
Nickel Sulphate + Chromium	Calcium Oxide + Lead	Cobalt hydroxide + Lead
Potassium Oxide + Sodium	Copper Chloride + Nickel	Iron Sulphate + Magnesium
Magnesium + Lead Oxide	Manganese Sulphate + Barium	Strontium Oxide + Copper

Displacement Reactions

Use the reactivity series to help you complete the word equations below.

Most Reactive

Metal	Symbol
Potassium	K
Sodium	Na
Lithium	Li
Barium	Ba
Strontium	Sr
Calcium	Ca
Magnesium	Mg
Aluminium	Al
Manganese	Mn
Zinc	Zn
Chromium	Cr
Iron	Fe
Cadmium	Cd
Cobalt	Co
Nickel	Ni
Tin	Sn
Lead	Pb
Copper	Cu
Silver	Ag
Platinum	Pt

Least Reactive

Calcium Oxide + Barium → Barium Oxide + Calcium	Sodium Chloride + Potassium → Potassium chloride + Sodium
Manganese Sulphate + Calcium → Calcium sulphate + Manganese	Iron Nitrate + Zinc → Zinc Nitrate + Iron
Tin Oxide + Aluminium → Aluminium Oxide + Tin	Lithium Oxide + Sodium → Sodium Oxide + Lithium
Cadmium Hydroxide + Aluminium → Aluminium Hydroxide + Cadmium	Silver Oxide + Copper → Copper Oxide + Silver
Strontium Sulphate + Potassium → Potassium Sulphate + Strontium	Cobalt Chloride + Iron → Iron Chloride + Cobalt
Tin Sulphate + Zinc → Zinc Sulphate + Tin	Platinum Chloride + Silver → Platinum Chloride + Silver

Colour all the reactions below that are not possible in green.

Zinc Oxide + Tin	Barium Sulphate + Lithium	Cobalt Chloride + Silver
Nickel Sulphate + Chromium	Calcium Oxide + Lead	Cobalt hydroxide + Lead
Potassium Oxide + Sodium	Copper Chloride + Nickel	Iron Sulphate + Magnesium
Magnesium + Lead Oxide	Manganese Sulphate + Barium	Strontium Oxide + Copper

Complete the worksheet

Displacement reactions:

- Watch the following videos:

https://www.youtube.com/watch?v=UIl0P3btVNI

https://www.youtube.com/watch?v=k8UtR7akNec

What observations can you make?

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